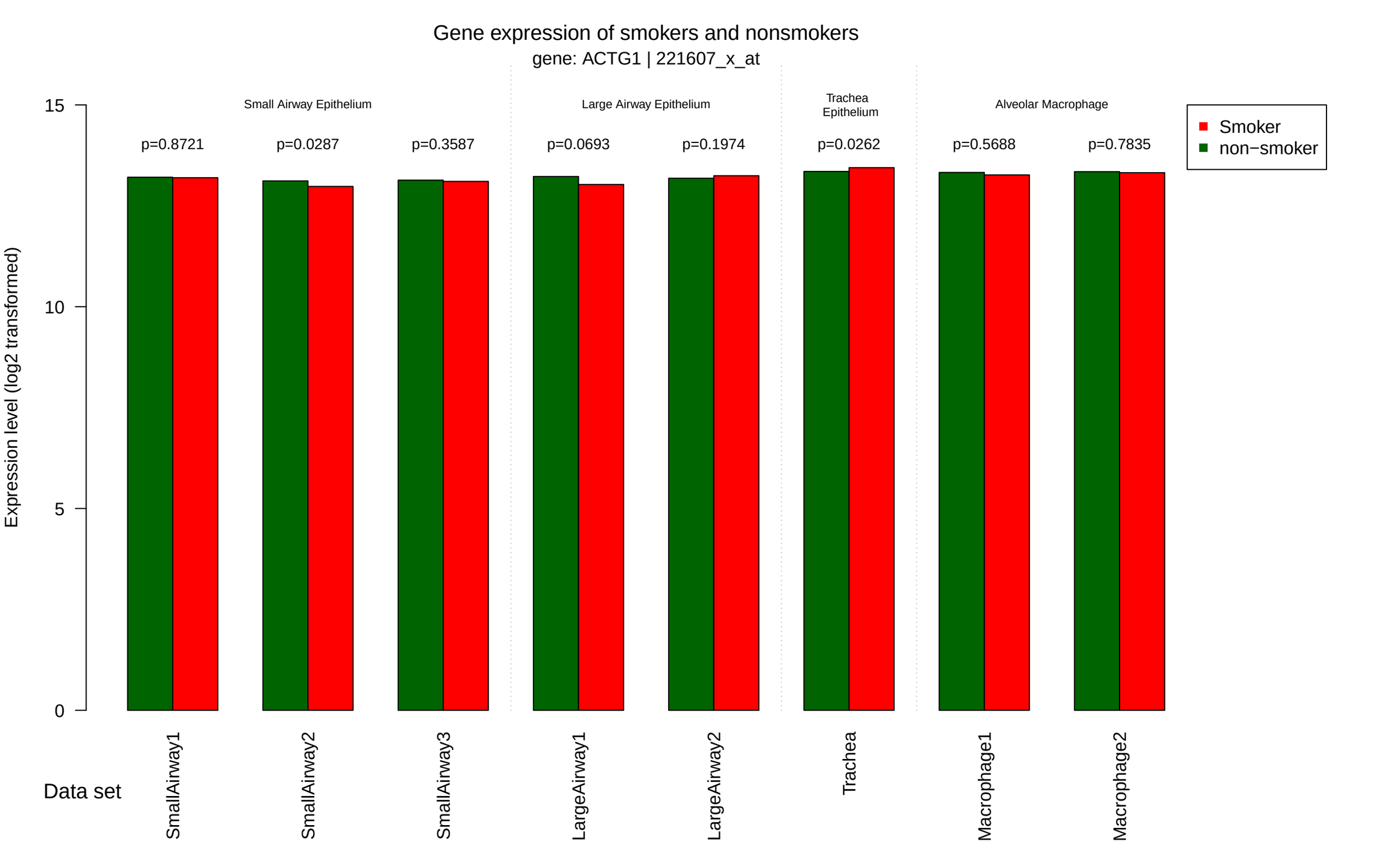
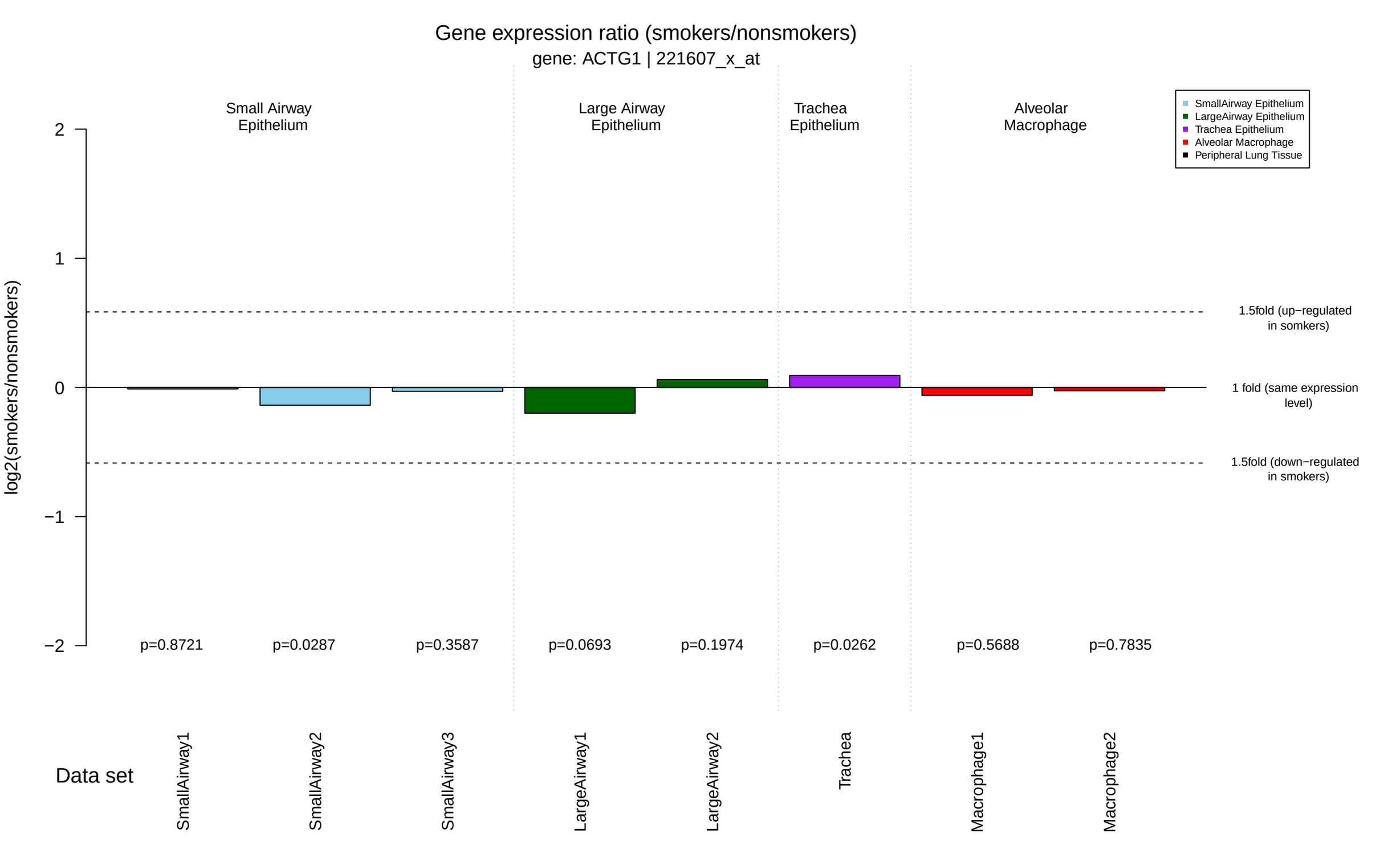
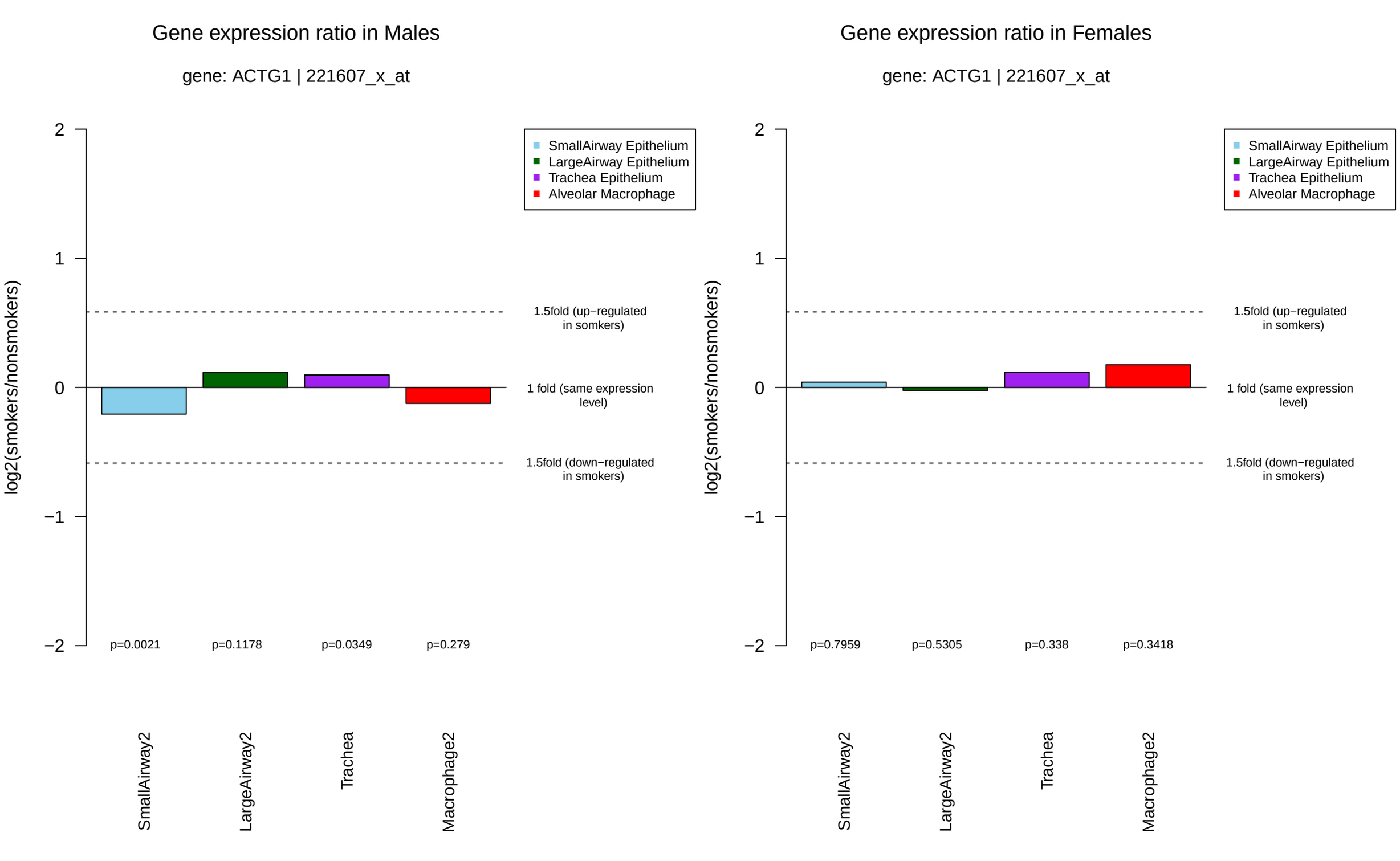
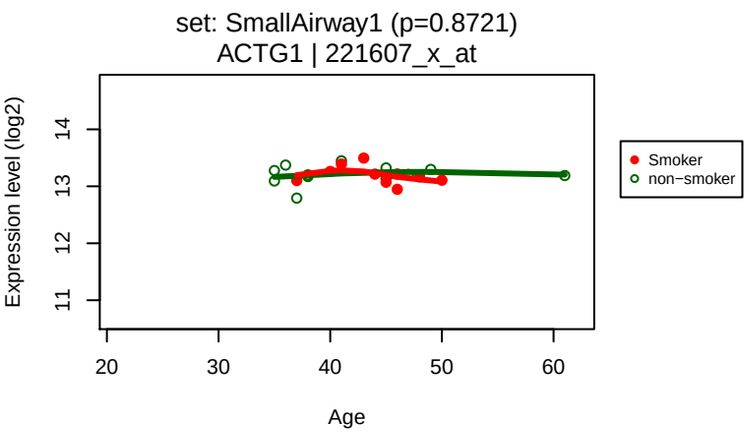
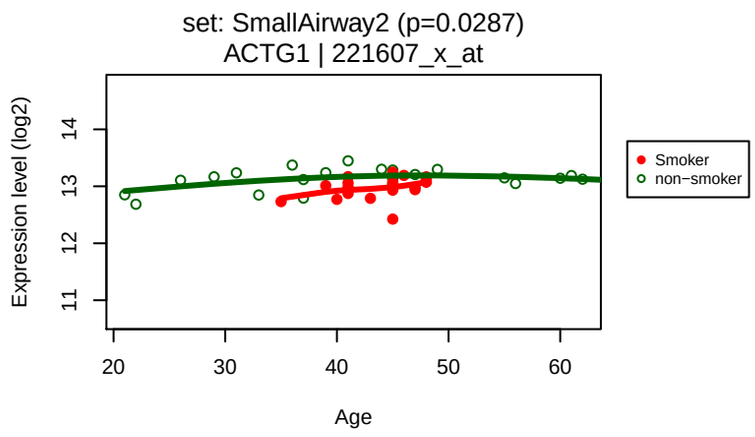
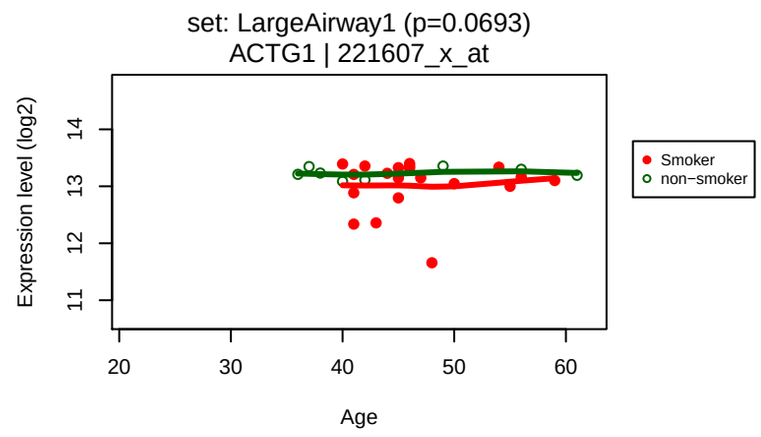

Gene expression of smokers and nonsmokers
gene: ACTG1 | 221607_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8721
p=0.0287
p=0.3587
p=0.0693
p=0.1974
p=0.0262
p=0.5688
p=0.7835
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

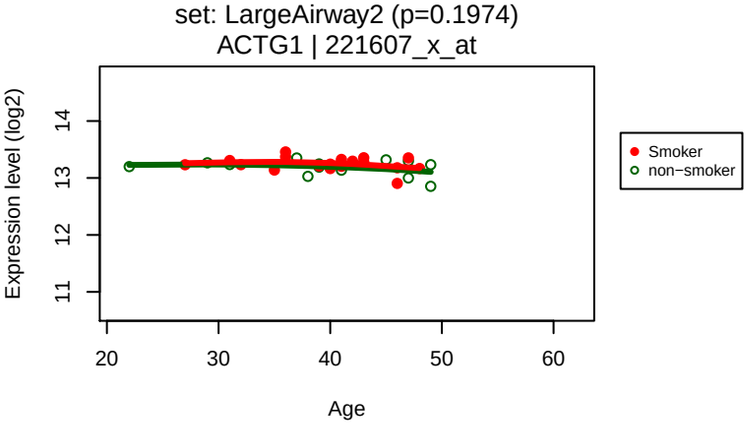

Gene expression ratio (smokers/nonsmokers)
gene: ACTG1 | 221607_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8721
p=0.0287
p=0.3587
p=0.0693
p=0.1974
p=0.0262
p=0.5688
p=0.7835
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

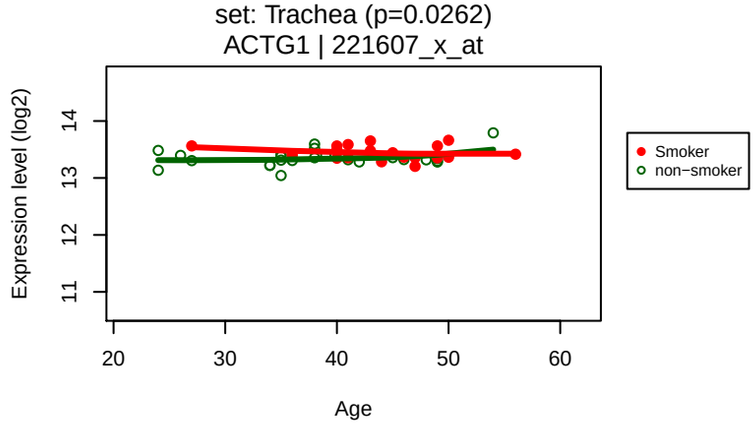

Gene expression ratio in Males
gene: ACTG1 | 221607_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0021
p=0.1178
p=0.0349
p=0.279
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTG1 | 221607_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7959
p=0.5305
p=0.338
p=0.3418
SmallAirway2
LargeAirway2
Trachea
Macrophage2

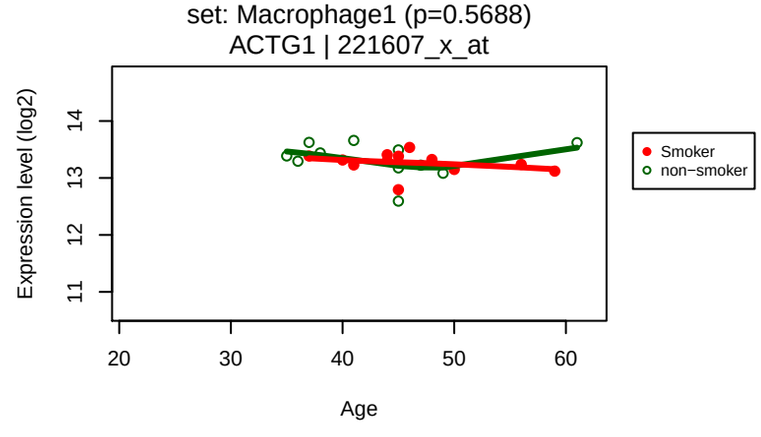

set: SmallAirway1 (p=0.8721)
ACTG1 | 221607_x_at
Expression level (log2)
Age
Smoker
non−smoker

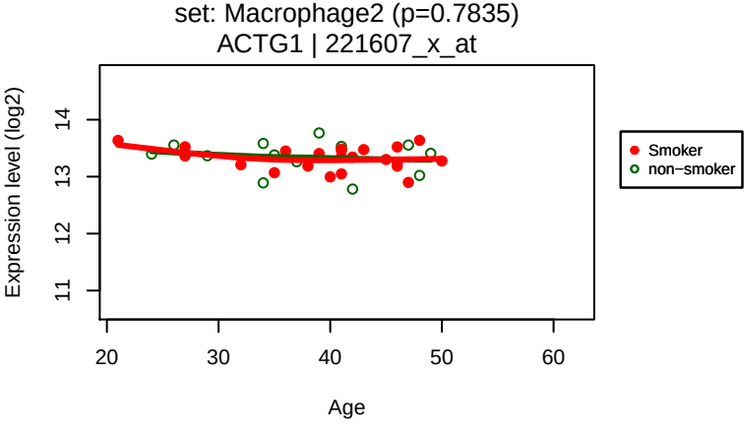

set: SmallAirway2 (p=0.0287)
ACTG1 | 221607_x_at
Expression level (log2)
Age
Smoker
non−smoker

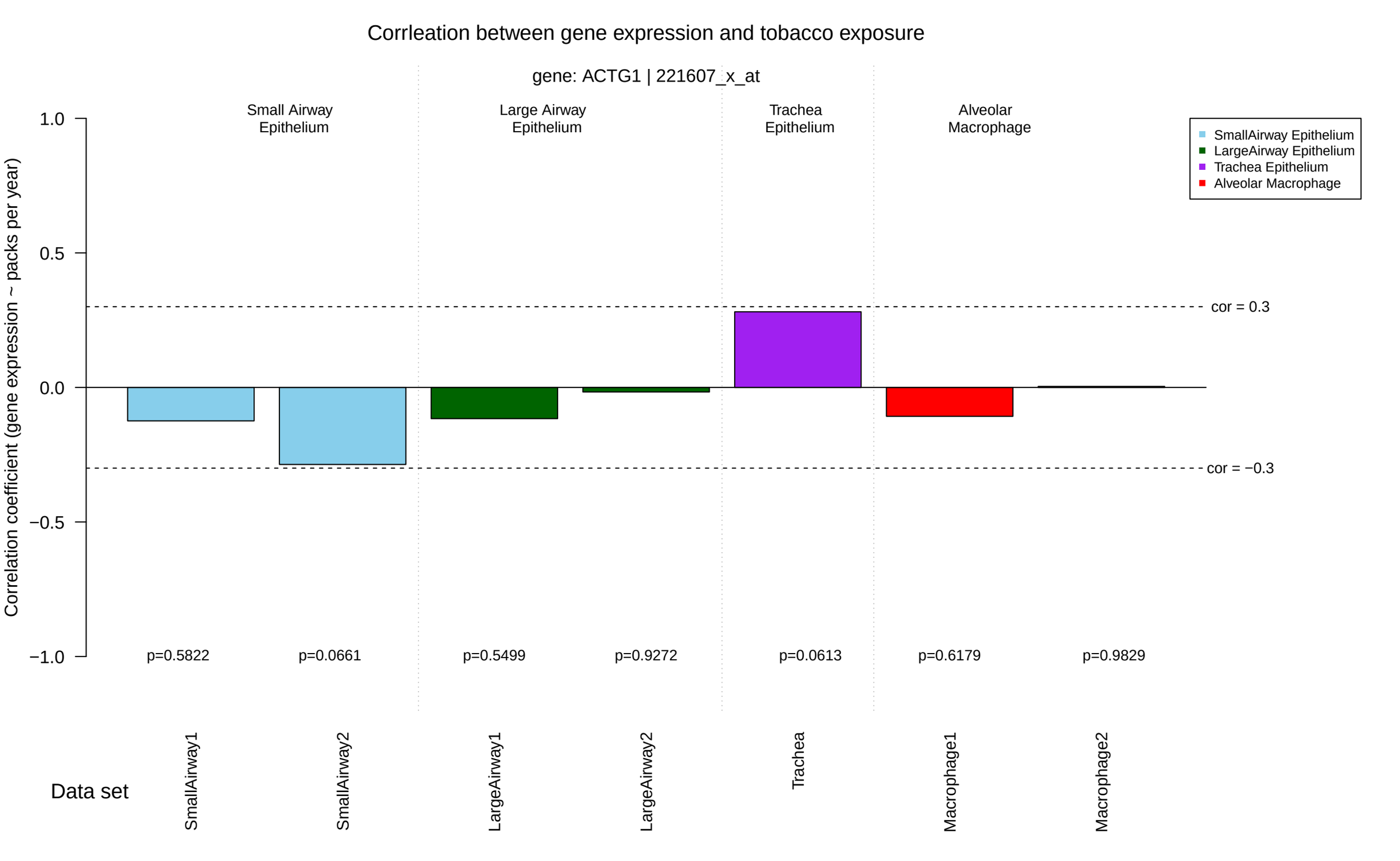

set: LargeAirway1 (p=0.0693)
ACTG1 | 221607_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1974)
ACTG1 | 221607_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0262)
ACTG1 | 221607_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5688)
ACTG1 | 221607_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7835)
ACTG1 | 221607_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTG1 | 221607_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5822
p=0.0661
p=0.5499
p=0.9272
p=0.0613
p=0.6179
p=0.9829
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2